A Course Mini Project report on

Job Scheduling System with Deadlines for Maximum Profit

Submitted in partial fulfilment of the requirements.

for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING (AI&ML)

by

Under the Guidance of

Ms. K. Rubiya Alam

Assistant Professor, H&SDepartment



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (ARITIFICAL INTELLIGENCE AND MACHINE LEARNING)

SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY (AUTONOMOUS) ANANTHAPURAMU

(Affiliated to JNTUA & Approved by AICTE) (Accredited by NAAC With 'A'Grade & Accredited by NBA (MEC,CIVEEE, ECE ,CAD,CSM& CSE))

2025-2026

VISION & MISSION

INSTITUTION VISION AND MISSION

VISION:

To become a premier Educational Institution in India offering the best teaching and learning environment for our students that will enable them to become complete individuals with professional competency, human touch, ethical values, service motto, and a strong sense of responsibility towards environment and society at large.

MISSION:

M1: Continually enhance the quality of physical infrastructure and human resources to evolve into a center of excellence in engineering education.

M2: Provide comprehensive learning experiences that are conducive for the students to acquire professional competences, ethical values, life-long learning abilities and understanding of the technology, environment and society.

M3: Strengthen industry institute interactions to enable the students work on realistic problems and acquire the ability to face the ever-changing requirements of the industry.

M4: Continually enhance the quality of the relationship between students and faculty which is a key to the development of an exciting and rewarding learning environment in the college.

DEPARTMENT VISION AND MISSION

VISION:

The vision of the Contact Management System is to deliver a robust, structure-based C application that revolutionizes contact tracking for small enterprises by enabling seamless digital organization, real-time updates, and effortless retrieval of contact details, ultimately eliminating manual errors and paperwork inefficiencies.

MISSION:

DM 1: Develop modular CRUD operations (Create, Read, Update, Delete) using structures and arrays to store up to 100 contacts efficiently.

DM 2: Implement a menu-driven interface with search, status updates, and basic reporting to simplify user interactions and reduce manual errors.

DM 3: Build modular C console system using structures for efficient CRUD,

SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY
(AUTONOMOUS) ANANTHAPURAMU
(Affiliated to JNTUA & Approved by AICTE) (Accredited by NAAC With 'A'Grade & Accredited by NBA (EEE, ECE & CSE))



Certificate

This is to certify that the Course Mini project report entitled by "Job Scheduling System with Deadlines for Maximum Profit" is the bona fide work carried out by **V. BHARATH REDDY** bearing Roll Number **254G1A3322**, in partial fulfilment of the requirements for the CAA during the academic year 2026-2027.

Guide **Head of the Department**

Ms. K.RUBIYA ALAM Assistant Professor,CSM Department.

DR. P. CHITRALINGAPPA Assistant Associate Professor & Head, CSMDepartment.

Date:

Place: Ananthapuramu

DECLARATION

We hereby declare that the course mini project report entitled "**Job scheduling System with Deadlines for Maximum Profit**" submitted for the CAA is our original work.

Place: Ananthapuramu

Date:

Course Mini Project Associates

V. BHARATH REDDY 254G1A3322

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of people who made it possible, whose constant guidance and encouragement crowned our efforts with success. It is a pleasant aspect that we have now the opportunity to express my gratitude for all of them.

It is with immense pleasure that we would like to express our indebted gratitude to our Guide **Ms. K .RUBIYA ALAM** Assistant professor H&S Department, who has guided us a lot and encouraged us in every step of the course mini project work. We thank him for the stimulating guidance, constant encouragement and constructive criticism which have made possible to bring out this course mini project work.

We are very much thankful to **Dr. P. CHITRALINGAPPA , HOD, Department of CSM ,** for his kind support and for providing necessary facilities to carry out the work.

We wish to convey my special thanks to **Dr. G. Balakrishna, Principal** of **Srinivasa Ramanujan Institute of Technology** for giving the required information in doing our course mini project work. Not to forget, we thank all other faculty and nonteaching staff, and our friends who had directly or indirectly helped and supported us in completing our course mini project in time.

We also express our sincere thanks to **Smt. J. Padmavathy, Chairperson** and **Sri A. Sambasiva Reddy, Secretary & Correspondent** of **Srinivasa Ramanujan Institute of Technology** for providing excellent facilities**.**

Finally, we wish to convey our gratitude to our family who fostered all the requirements and facilities that we need.

Table of Contents

S.no	Contents	Page.no
1	Abstract	7
2	Introduction	7
3	Objectives	8
4	Job Scheduling Program	8
5	Problem Definition	9
6	Basic Concepts	9
7	Greedy Algorithm	10
8	Job Scheduling With Deadlines	10
9	Proposed System	10
10	Methodology	11
11	Algorithm	11
12	Implementation	13
13	Results and Analysis	13
14	Advantages	14
15	Applications	14
16	Limitations	14
17	Future Scope	15
18	Conclusion	15
19	References	16

ABSTRACT

Job Scheduling with Deadlines is an important optimization problem in computer science and is commonly solved using the **Greedy Algorithm**. In this problem, a set of jobs is given where every job has a deadline and a profit. Each job requires one unit of time for execution and must be completed before or on its deadline to obtain its profit.

The main objective of the project is to select and schedule jobs in such a way that the **maximum possible total profit** is obtained while satisfying all job deadlines.

The proposed Job Scheduling System accepts job details such as job ID, deadline, and profit. The jobs are arranged according to their profit in descending order. Each job is then assigned to the latest available time slot before its deadline. If a suitable slot is available, the job is scheduled; otherwise, it is skipped.

This project demonstrates the practical application of greedy problem-solving techniques and helps students understand optimization, scheduling, sorting, and algorithmic decision-making. The system can be applied to task management, production scheduling, project planning, and resource allocation.

INTRODUCTION

In many real-world systems, several jobs or tasks need to be completed within specific deadlines. Each task may provide a different amount of profit or benefit. Since available time is limited, it is not always possible to execute every job.

The **Job Scheduling with Deadlines** problem focuses on selecting a suitable set of jobs so that each selected job is completed within its deadline and the total profit is maximized.

For example, consider a company having several tasks. Each task has a deadline and a payment. Since only one task can be completed during a particular time slot, the company needs to decide which tasks should be performed and in what order.

A greedy approach provides an efficient solution for the standard version of this problem. The algorithm first considers jobs with higher profits and attempts to place each job in the latest available slot before its deadline.

This project implements the concept as a simple Job Scheduling System and demonstrates how algorithms can be used to solve practical optimization problems.

.

OBJECTIVES

The main objectives of this mini project are:

- To understand the Job Scheduling with Deadlines problem.
- To understand the concept of greedy algorithms.
- To schedule jobs according to their deadlines.
- To maximize the total profit obtained from selected jobs.
- To implement sorting based on job profits.
- To identify available time slots for each job.
- To develop a simple and efficient job scheduling system.
- To analyze the time and space complexity of the algorithm.
- To understand practical applications of job scheduling.
- To improve algorithmic problem-solving skills.

JOB SCHEDULING PROGRAM

The Job Scheduling Problem is an optimization problem in which a collection of jobs is given. Every job has:

1. **Job ID**
2. **Deadline**
3. **Profit**

Each job requires exactly one unit of time to complete.

The objective is to select jobs that can be completed within their deadlines while obtaining the maximum possible total profit.

Example

Job	**Deadline**	**Profit**
J1	2	100
J2	1	19
J3	2	27
J4	1	25
J5	3	15

The system must determine which jobs should be selected and their order of execution.

PROBLEM DEFINITION

Given **N jobs**, where every job has a deadline and a profit, schedule a subset of jobs such that:

- Every selected job is completed before its deadline.
- Only one job is executed in each time slot.
- The total profit of selected jobs is maximized.

Input

For every job:

- Job ID
- Deadline
- Profit

Output

- Selected jobs
- Scheduled order
- Time slots
- Maximum total profit

BASIC CONCEPTS

Job

A job represents a task that needs to be completed.

Deadline

The deadline represents the latest time by which a job must be completed to receive its profit.

Profit

Profit is the benefit obtained when a job is successfully completed within its deadline.

Time Slot

A time slot represents one unit of available execution time. Since every job requires one unit of time, only one job can occupy a particular slot.

Scheduling

Scheduling is the process of assigning jobs to available time slots while satisfying their constraints.

Optimization

Optimization means finding a solution that provides the maximum possible total profit.

GREEDY ALGORITHM

A greedy algorithm solves a problem by making the best available choice at every step.

For Job Scheduling with Deadlines, the greedy strategy is:

1. Sort all jobs in descending order of profit.
2. Consider the highest-profit job first.
3. Find the latest available slot before its deadline.
4. Schedule the job if a slot is available.
5. Continue until all jobs are considered.

The reason for using the latest available slot is to keep earlier slots available for jobs having smaller deadlines

JOB SCHEDULING WITH DEADLINES

The scheduling process can be represented as:

Input Jobs → Sort by Profit → Check Deadline → Find Free Slot → Schedule Job → Calculate Total Profit

Main components

Job List: Stores all jobs and their information.

Sorting Module: Arranges jobs according to profit.

Slot Allocation Module: Searches for an available slot.

Profit Calculation Module: Calculates the total profit obtained.

Output Module: Displays the selected jobs and maximum profit**.**

PROPOSED SYSTEM

The proposed system provides a simple mechanism for solving the job scheduling problem.

System Workflow

1. Start the program.
2. Enter the number of jobs.
3. Enter job ID, deadline, and profit.
4. Sort jobs according to profit in descending order.
5. Determine the maximum deadline.
6. Create available time slots.
7. Select the highest-profit job.

8. Search for the latest free slot before its deadline.
9. Assign the job to that slot.
10. Repeat for remaining jobs.
11. Calculate total profit.
12. Display the final schedule.
13. Stop.

METHODOLOGY

The project follows the following methodology.

Step 1: Input

The user provides job details.

Step 2: Sorting

All jobs are sorted in descending order of profit.

Step 3: Slot Creation

Time slots are created based on the maximum deadline.

Step 4: Job Selection

Jobs are considered one by one according to their profit.

Step 5: Slot Allocation

For each job, the system searches backward from its deadline for an empty slot.

Step 6: Profit Calculation

The profit of every successfully scheduled job is added.

Step 7: Output

The system displays the final job sequence and maximum profit.

ALGORITHM

Algorithm: Job Scheduling with Deadlines

Input: Set of jobs with deadlines and profits.

Output: Schedule of jobs with maximum profit.

1. Start.
2. Read the number of jobs.
3. Read job ID, deadline, and profit for each job.
4. Sort jobs in decreasing order of profit.
5. Find the maximum deadline.
6. Create time slots and mark all slots as empty.

7. For each job:
    - Start from its deadline.
    - Search backwards for an empty slot.
    - If an empty slot is found, assign the job.
    - Add its profit to the total profit.
8. Display the scheduled jobs.
9. Display the maximum total profit.
10. Stop.

EXAMPLE

Consider the following jobs:

Job	Deadline	Profit
J1	2	100
J2	1	19
J3	2	27
J4	1	25
J5	3	15

Step 1: Sort according to profit

J1 → 100
J3 → 27
J4 → 25
J2 → 19
J5 → 15

Step 2: Allocate slots

There are three available slots because the maximum deadline is 3.

- J1 → Slot 2
- J3 → Slot 1
- J4 → No free slot before deadline 1
- J2 → No free slot before deadline 1
- J5 → Slot 3

Final Schedule

Time Slot	Job	Profit
1	J3	27
2	J1	100
3	J5	15

Maximum Profit
27 + 100 + 15 = 142
Therefore, the maximum profit obtained is **142**.

IMPLEMENTATION

The project can be implemented using **C programming language**.
Basic data structure

```
struct Job {
char id[10];
int deadline;
int profit;
 };
```

The program stores the details of every job using a structure. Sorting is performed based profit, and an array is used to represent available time slots.

Important operations

- Reading job information
- Sorting jobs
- Finding available slots
- Scheduling jobs
- Calculating total profit
- Displaying result

RESULTS AND ANALYSIS

The Job Scheduling System successfully selects jobs according to their profit and deadlines.

For the sample input:
Jobs: J1, J2, J3, J4, J5
Selected Jobs: J3, J1, J5
Total Maximum Profit: 142
The result demonstrates that the greedy approach can efficiently identify a profitable feasible schedule for the standard unit-time job scheduling problem.

Time Complexity
Sorting the jobs requires approximately**:**
O (N log N)

Slot allocation can require up **to:**
O (N × D)
where N is the number of jobs and D is the maximum deadline**.**

Space Complexity
The space required for storing jobs and slots is approximately:
O (N + D)

ADVANTAGES

- Simple and easy to understand.
- Efficient for the standard job scheduling problem.
- Maximizes total profit under the given constraints.
- Uses a straightforward greedy strategy.
- Requires relatively low memory.
- Can handle multiple jobs.
- Useful for understanding optimization algorithms.
- Can be implemented using basic data structures.
- Suitable for academic demonstrations.
- Can be extended to practical scheduling applications.

APPLICATIONS

Job scheduling concepts can be applied in:

1. **Project Management** – Selecting profitable tasks within deadlines.
2. **Task Scheduling** – Managing multiple computer or organizational tasks.
3. **Production Planning** – Selecting manufacturing jobs.
4. **Business Management** – Prioritizing profitable activities.
5. **Resource Allocation** – Assigning limited resources to tasks.
6. **Computer Systems** – Scheduling computational jobs.
7. **Service Management** – Prioritizing customer tasks.
8. **Freelancing Platforms** – Selecting jobs based on payment and deadlines.

LIMITATIONS

- The basic algorithm assumes that every job requires one unit of time.
- It does not consider different execution times.
- It assumes that jobs are independent.
- The standard approach focuses on profit and deadline constraints.
- Real-world scheduling may involve additional constraints such as resources, priorities, dependencies, and interruptions

FUTURE SCOPE

The system can be improved by adding:

- Different processing times for different jobs.
- Multiple machines or processors.
- Job priorities.
- Resource constraints.
- User-friendly graphical interface.
- Database storage for job information.
- Dynamic job insertion.
- Real-time scheduling.
- Deadline notifications.
- Performance comparison between different scheduling algorithms.
- Web or mobile-based job management.

CONCLUSION

The **Job Scheduling System with Deadlines for Maximum Profit** demonstrates how an optimization problem can be solved using a greedy algorithm. The system considers a set of jobs, each having a deadline and profit, and selects a feasible sequence of jobs that maximizes the total profit.

The algorithm first sorts the jobs according to their profit and then attempts to place each job in the latest available time slot before its deadline. This approach provides an efficient solution for the standard unit-time job scheduling problem.

Through this project, important concepts such as **greedy algorithms, sorting, scheduling, optimization, time complexity, and space complexity** are understood practically.

The project also demonstrates how theoretical concepts from data structures and algorithms can be applied to real-world scheduling and resource-allocation problems. With further development, the system can support additional constraints and be integrated into larger task-management and scheduling applications.

.

REFERENCES

1. **GeeksforGeeks – Job Sequencing Problem**
   Covers the job sequencing problem, deadlines, profits, greedy approach, and complexity.
   Job Sequencing Problem – GeeksforGeeks
2. **GeeksforGeeks – Job Sequencing Problem using Disjoint Set**
   Explains an optimized approach using the Disjoint Set/Union-Find technique.
   Job Sequencing using Disjoint Set – GeeksforGeeks
3. **GeeksforGeeks – Job Sequencing Problem Practice**
   Provides problem statements, examples, constraints, and expected complexity for implementation practice.
   Job Sequencing Problem – Practice
4. **TutorialsPoint – Job Sequencing with Deadline**
   Provides the greedy algorithm, step-by-step example, implementation, and maximum-profit calculation.
   Job Sequencing with Deadline – TutorialsPoint
5. **TutorialsPoint – Quiz on Job Sequencing with Deadline**
   Useful for understanding and practicing the concepts related to job scheduling with deadlines.
   Job Sequencing with Deadline Quiz – TutorialsPoint
6. **Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. – Introduction to Algorithms**
   A standard reference for algorithms, including greedy algorithm concepts and algorithm analysis.
7. **Horowitz, E., Sahni, S., & Rajasekaran, S. – Computer Algorithms**
   A useful academic reference for algorithm design, optimization, and scheduling techniques..

PO's & PSO's

PO 1: Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design / Development Of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct Investigations Of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The Engineer And Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment And Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual And Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project Management And Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-Long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes

At the end of the B. Tech program in Computer Science and Engineering(Data Science), the graduate will be able to:

PSO1: Use the concepts and practical knowledge in analysis, design, implement and development of computing systems and applications to multi-disciplinary problems.

PSO2: Design information systems using standard statistical tools and techniques to analyze large data for visualization and interpretation.

PSO3: Expertise in developing relevant Data science models and working with extensive data sets required to meet the industry's growing demand for data scientists and engineers

Evaluation Criteria	Maximum Marks	Marks Obtained
Problem Understanding & Clarity	2	
Use of Aids (Figures/Charts/Models)	2	
Application of Course Concepts	2	
Methodology & Analysis	2	

Results, Findings & Discussion	2	
Total	**10**	

Evaluation Parameters

Faculty Feedback / Remarks

SIGNATURE OF FACULITY **SIGNATURE OF STUDENT**